Profile comparison

1 profiles from C:\APPS\FBENCH\

Machines

	Name / file	Screen	Planes	Test window	Desktop	Blitter	Date
A	**NewTOS on Raspberry PI 4**	1920x1080	16	1276x997	1920x1061	none	2026-11-06 00..
	MACHINE.FBP - unknown AArch64 TOS 4.04 1920x1080 256c bli						

Machine A (NewTOS on Raspberry PI 4) is the reference: percentages are that machine's speed as 100%. Bold green marks the best machine for each test.

Summary - raw speed (% of A)

Average of the per-test percentages in each group, exactly as measured. A machine running a smaller window or fewer colours does less work per operation and looks faster here.



Summary - normalised (% of A)

The same averages after adjusting each test for the work one operation does on that machine: colour depth (bit planes), the size of the test window, and the desktop size for full-screen tests. This compares graphics throughput independent of the screen mode.



Raw results (operations per second)

VDI primitives

	A
Line, 2 points	22414
Polyline, 32 points	763.1
Wide line (7px, round)	3327.0
Dashed line	20572
User line style	21765
XOR line	11263

Markers x16	22754
Filled rect 32x32	37235
Filled rect, full window	35.4
Bar, hatched + border	21977
Pattern fill rect	37750
User pattern fill	37925
XOR filled rect	28108
Polygon, 8 points	13985
Filled circle	682.5
Filled ellipse	892.0
Arc	20858
Pie slice	765.8
Elliptical arc	21870
Elliptical pie	1007.8
Rounded box	39766
Filled rounded box	3168.5
Contour fill 64x40	4265.7
Text, 15 chars	17225
Text, 84 chars	2967.1
Text with effects	7455.3
Text rotated 90	15905
Justified text	10477
Text extent query	383624
Blit scr->scr 64x64	22414
Blit scroll window	119.2
Blit mem->scr 64x64	27971
Blit scr->mem 64x64	24765
Mono blit (vrt_cpyfm)	15282
Get pixel	481881
Hide/show mouse	20858
Attribute calls x3	165342
Open/close workstation	46151

AES windows & events

	A
Window create+delete	135279
Window open+close	26.2
Window move	335.8
Window resize	198.0
Window top (2 windows)	52.1
Set window title	860.5
Set info line	864.7
Set slider pos+size	1247.8
Scroll window content	112.7
wind_get work area	338447
Rectangle list walk	148270
wind_calc	280867
wind_update pair	169223
objc_draw dialog	3621.6
objc_find	287718
form_dial redraw 64x48	4305.6
Menu bar redraw	3975.7
graf_mkstate	340044
graf_mouse off+on	19786
Event poll (evnt_multi)	137204
Message round trip	61535

Alignment (blitter)

	A
Fill 64x64, aligned	11376
Fill 64x64, x+3	10934
Blit 64x64, aligned	24093
Blit 64x64, src+3 dst+11	21152
Mem->scr 64x64, aligned	28108
Mem->scr 64x64, x+3	27835
Mono blit, aligned	15282
Mono blit, x+3	15132
Text, x on 8 px	17981
Text, x+3	16383
Scroll up, aligned edges	123.9
Scroll up, edges x+5	123.9
Scroll left 16 px	124.5
Scroll left 3 px	123.9
Move window, aligned	320.5
Move window, off-grid	317.4
Open/close win, aligned	25.9
Open/close win, off-grid	25.9
Scroll window, aligned	347.5
Scroll window, off-grid	342.4

Full-screen windows

	A
Open/close full window	13.1
Full <-> normal size	26.0
Top full / test window	26.7
Fill full window	23.3
Scroll up 1 line	77.4
Scroll left 16 px	76.6
Scroll left 8 px	79.0
Scroll left 3 px	79.8
Scroll text 8pt	80.1
Scroll text 10pt	77.4
Scroll text 12pt	77.4
Scroll text 14pt	77.4
Scroll text 18pt	76.6
Scroll text 24pt	72.4
Scroll text 36pt	72.4
Scroll text 48pt	72.4
Scroll, size per line	75.5

GDOS fonts

	A
Text 6pt	37066
Text 8pt	37066
Text 10pt	12986
Text 12pt	12923
Text 14pt	12986
Text 18pt	6761.3
Text 24pt	3478.6
Text 36pt	3495.6
Text 48pt	3512.7
Text 72pt	3461.8
Text 10pt bold	12737
vst_point, 8 sizes	381762
Scroll, mixed sizes	108.2
Text extent at 12pt	383624

Relative to A (raw, %)

VDI primitives

	A
Line, 2 points	100%
Polyline, 32 points	100%
Wide line (7px, round)	100%
Dashed line	100%
User line style	100%
XOR line	100%
Markers x16	100%
Filled rect 32x32	100%
Filled rect, full window	100%
Bar, hatched + border	100%
Pattern fill rect	100%
User pattern fill	100%
XOR filled rect	100%
Polygon, 8 points	100%
Filled circle	100%
Filled ellipse	100%
Arc	100%
Pie slice	100%
Elliptical arc	100%
Elliptical pie	100%
Rounded box	100%
Filled rounded box	100%
Contour fill 64x40	100%
Text, 15 chars	100%
Text, 84 chars	100%
Text with effects	100%
Text rotated 90	100%
Justified text	100%
Text extent query	100%
Blit scr->scr 64x64	100%
Blit scroll window	100%
Blit mem->scr 64x64	100%
Blit scr->mem 64x64	100%
Mono blit (vrt_cpyfm)	100%
Get pixel	100%
Hide/show mouse	100%
Attribute calls x3	100%
Open/close workstation	100%

AES windows & events

	A
Window create+delete	100%
Window open+close	100%
Window move	100%
Window resize	100%
Window top (2 windows)	100%
Set window title	100%
Set info line	100%
Set slider pos+size	100%
Scroll window content	100%
wind_get work area	100%
Rectangle list walk	100%
wind_calc	100%
wind_update pair	100%
objc_draw dialog	100%

objc_find	100%
form_dial redraw 64x48	100%
Menu bar redraw	100%
graf_mkstate	100%
graf_mouse off+on	100%
Event poll (evnt_multi)	100%
Message round trip	100%

Alignment (blitter)



Fill 64x64, aligned	100%
Fill 64x64, x+3	100%
Blit 64x64, aligned	100%
Blit 64x64, src+3 dst+11	100%
Mem->scr 64x64, aligned	100%
Mem->scr 64x64, x+3	100%
Mono blit, aligned	100%
Mono blit, x+3	100%
Text, x on 8 px	100%
Text, x+3	100%
Scroll up, aligned edges	100%
Scroll up, edges x+5	100%
Scroll left 16 px	100%
Scroll left 3 px	100%
Move window, aligned	100%
Move window, off-grid	100%
Open/close win, aligned	100%
Open/close win, off-grid	100%
Scroll window, aligned	100%
Scroll window, off-grid	100%

Full-screen windows



Open/close full window	100%
Full <-> normal size	100%
Top full / test window	100%
Fill full window	100%
Scroll up 1 line	100%
Scroll left 16 px	100%
Scroll left 8 px	100%
Scroll left 3 px	100%
Scroll text 8pt	100%
Scroll text 10pt	100%
Scroll text 12pt	100%
Scroll text 14pt	100%
Scroll text 18pt	100%
Scroll text 24pt	100%
Scroll text 36pt	100%
Scroll text 48pt	100%
Scroll, size per line	100%

GDOS fonts



Text 6pt	100%
Text 8pt	100%
Text 10pt	100%
Text 12pt	100%
Text 14pt	100%
Text 18pt	100%
Text 24pt	100%

Text 36pt	100%
Text 48pt	100%
Text 72pt	100%
Text 10pt bold	100%
vst_point, 8 sizes	100%
Scroll, mixed sizes	100%
Text extent at 12pt	100%

Normalised results (% of A)

Each test's operations per second are multiplied by the work one operation does on that machine, then shown as a percentage of machine A. The 'scales' column says what the work is proportional to:

planes - fixed-size drawing (64x64 blits, text, 32x32 fills): bit planes only. len - lines across the window: (width + height) / 2 x planes. min - arcs and rounded boxes sized from the window: min(width, height) x planes. dim² - filled shapes whose radius follows the window: min(w, h)² x planes. area - whole test-window work (fills, scrolls, window moves): width x height x planes. desk - full-screen windows: desktop width x height x planes. calls - AES/VDI calls with no pixel work (events, messages, queries): not adjusted.

VDI primitives

	scales	A
Line, 2 points	len	100%
Polyline, 32 points	len	100%
Wide line (7px, round)	len	100%
Dashed line	len	100%
User line style	len	100%
XOR line	len	100%
Markers x16	planes	100%
Filled rect 32x32	planes	100%
Filled rect, full window	area	100%
Bar, hatched + border	planes	100%
Pattern fill rect	planes	100%
User pattern fill	planes	100%
XOR filled rect	planes	100%
Polygon, 8 points	planes	100%
Filled circle	dim²	100%
Filled ellipse	dim²	100%
Arc	min	100%
Pie slice	dim²	100%
Elliptical arc	min	100%
Elliptical pie	dim²	100%
Rounded box	min	100%
Filled rounded box	dim²	100%
Contour fill 64x40	planes	100%
Text, 15 chars	planes	100%
Text, 84 chars	planes	100%
Text with effects	planes	100%
Text rotated 90	planes	100%
Justified text	planes	100%
Text extent query	calls	100%
Blit scr->scr 64x64	planes	100%
Blit scroll window	area	100%
Blit mem->scr 64x64	planes	100%
Blit scr->mem 64x64	planes	100%
Mono blit (vrt_cpyfm)	planes	100%
Get pixel	calls	100%
Hide/show mouse	calls	100%
Attribute calls x3	calls	100%
Open/close workstation	calls	100%

AES windows & events

	scales	A
Window create+delete	calls	100%
Window open+close	area	100%
Window move	area	100%
Window resize	area	100%
Window top (2 windows)	area	100%
Set window title	calls	100%
Set info line	calls	100%
Set slider pos+size	calls	100%
Scroll window content	area	100%
wind_get work area	calls	100%
Rectangle list walk	calls	100%
wind_calc	calls	100%
wind_update pair	calls	100%
objc_draw dialog	planes	100%
objc_find	calls	100%
form_dial redraw 64x48	planes	100%
Menu bar redraw	calls	100%
graf_mkstate	calls	100%
graf_mouse off+on	calls	100%
Event poll (evnt_multi)	calls	100%
Message round trip	calls	100%

Alignment (blitter)

	scales	A
Fill 64x64, aligned	planes	100%
Fill 64x64, x+3	planes	100%
Blit 64x64, aligned	planes	100%
Blit 64x64, src+3 dst+11	planes	100%
Mem->scr 64x64, aligned	planes	100%
Mem->scr 64x64, x+3	planes	100%
Mono blit, aligned	planes	100%
Mono blit, x+3	planes	100%
Text, x on 8 px	planes	100%
Text, x+3	planes	100%
Scroll up, aligned edges	area	100%
Scroll up, edges x+5	area	100%
Scroll left 16 px	area	100%
Scroll left 3 px	area	100%
Move window, aligned	area	100%
Move window, off-grid	area	100%
Open/close win, aligned	area	100%
Open/close win, off-grid	area	100%
Scroll window, aligned	area	100%
Scroll window, off-grid	area	100%

Full-screen windows

	scales	A
Open/close full window	desk	100%
Full <-> normal size	desk	100%
Top full / test window	desk	100%
Fill full window	desk	100%
Scroll up 1 line	desk	100%
Scroll left 16 px	desk	100%
Scroll left 8 px	desk	100%
Scroll left 3 px	desk	100%
Scroll text 8pt	desk	100%
Scroll text 10pt	desk	100%

Scroll text 12pt	desk	100%
Scroll text 14pt	desk	100%
Scroll text 18pt	desk	100%
Scroll text 24pt	desk	100%
Scroll text 36pt	desk	100%
Scroll text 48pt	desk	100%
Scroll, size per line	desk	100%

GDOS fonts

	scales	A
Text 6pt	planes	100%
Text 8pt	planes	100%
Text 10pt	planes	100%
Text 12pt	planes	100%
Text 14pt	planes	100%
Text 18pt	planes	100%
Text 24pt	planes	100%
Text 36pt	planes	100%
Text 48pt	planes	100%
Text 72pt	planes	100%
Text 10pt bold	planes	100%
vst_point, 8 sizes	calls	100%
Scroll, mixed sizes	area	100%
Text extent at 12pt	calls	100%

Notes

Normalisation is a model, not a measurement: it assumes drawing cost grows with the number of bit planes touched and the pixels covered. Real hardware adds fixed per-call overheads, so small operations on high-resolution modes are usually under-rewarded and large ones fit best. GDOS text at the same point size covers different pixel counts at different screen resolutions (DPI); that is not adjusted. Profiles measured with different test durations, blitter settings or background programs are not directly comparable.